TopoRefiner: Universal Post-hoc Topology Refinement for Curvilinear Structures

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Abstract

Modern segmentation models can be locally accurate yet globally wrong: a few pixels may disconnect an entire structure or introduce a false connection while barely affecting overlap metrics. We introduce TopoRefiner, a post-hoc framework that learns structural correction in a topology-sensitive latent space. Given an image, a frozen segmentation, and its confidence field, TopoRefiner predicts a corrected structural representation and renders the resulting edit back into segmentation space without modifying or retraining the upstream model. Rather than tying topology correction to a specific architecture or loss, TopoRefiner treats it as a reusable post-hoc capability. We evaluate specialist, shared, and rollout-supervised variants across heterogeneous 2D road-segmentation systems and 3D medical-image segmentation benchmarks, including topology-aware predictors and existing refiners. Across these settings, TopoRefiner consistently improves structural fidelity while largely preserving segmentation quality, demonstrating that topology correction can be learned as a transferable operation across segmentation systems and domains.

1. Introduction

A segmentation can be nearly correct everywhere and still be fundamentally wrong as a structure. A gap only a few pixels wide can disconnect an otherwise continuous road, interrupt a surface crack, or fragment a thin anatomical structure. Conversely, a short false bridge can merge unrelated components, while a false branch can alter the connectivity of the recovered network. Such errors may barely affect region overlap while substantially changing the structural interpretation of the prediction. For applications involving mapping, navigation, structural inspection, or the analysis of interconnected structures, pixel accuracy and topological correctness are therefore not interchangeable.

Topology-aware losses and connectivity-driven architectures address this problem during segmentation [5, 6, 10, 16–18]. By incorporating structural constraints into the training objective, these approaches encourage the underlying predictor to preserve connected components, branches, and cycles. However, improving topology through the segmentation objective generally requires modifying or retraining the model that produces the prediction. Post-hoc refinement offers an attractive alternative because it can improve structural quality without changing the original segmentation model. Existing approaches range from generic mask refinement [4, 21, 23] to topology-oriented correction [3, 7, 13, 14]. UTR [14], for example, learns model-agnostic refinement from synthetic topological perturbations, while Curvi-Tracker [7] restores curvilinear connectivity through explicit image-guided tracing. Yet these methods either operate directly in segmentation space, rely on prescribed corruption processes, or reconstruct missing structures through sequential local decisions. They do not explicitly learn structural correction as a transformation in a topology-sensitive representation.

We introduce **TopoRefiner**, a lightweight post-hoc framework that learns precisely such a transformation. TopoRefiner first constructs a latent space in which topology-changing perturbations induce stronger representation changes than approximately topology-preserving geometric variations. Given an image, a frozen upstream segmentation, and its confidence field, a latent refiner predicts a corrected structural representation; the resulting latent edit is then rendered back into segmentation space. The upstream model remains completely frozen. Consequently, TopoRefiner can be applied after conventional segmenters, predictors trained with topology-aware objectives, and existing post-hoc refinement methods without modifying any of them. A central property of the framework is that the correction mechanism is not tied to a particular upstream architecture. We therefore study both **specialist** variants trained for a single predictor and **shared** variants trained across heterogeneous frozen predictors without model identities or architecture-specific features. We further study **rollout-**

supervised counterparts that learn to perform repeated structural correction. These variants probe complementary questions—specialization versus transferability, and single-step versus iterative correction—while sharing the same underlying TopoRefiner formulation. We evaluate TopoRefiner across three 2D road-segmentation benchmarks and two 3D medical-image segmentation benchmarks, using upstream systems that include conventional segmentation models, topology-aware training objectives, and existing topology refiners. Across these settings, TopoRefiner consistently provides substantial structural improvements while exposing an important trade-off between geometric fidelity and topological correction: the variant that best preserves overlap is not always the one that produces the strongest topological improvement. This behavior motivates evaluating topology correction directly rather than relying on overlap metrics alone. We additionally analyze structural disconnections to distinguish successful repairs from partial or incorrect modifications. Our main contributions are:

- We introduce TopoRefiner, a post-hoc topology-refinement framework that learns structural corrections in a topology-sensitive latent space and renders them back into segmentation space without modifying or retraining the upstream model.
- We formulate topology correction as a reusable operation across segmentation systems, and study specialist, shared, and rollout-supervised variants to characterize the effects of cross-model transfer and repeated structural correction.
- We provide a broad evaluation across 2D road segmentation and 3D medical-image segmentation, including topology-aware predictors and existing post-hoc refiners, together with direct analysis of connectivity repair and the trade-off between structural correctness and geometric fidelity.

2. Related Work

2.1. Curvilinear Structure Extraction and Connectivity

Extracting thin, interconnected structures from images requires more than accurate pixel classification: small discontinuities can break road networks or fragment surface cracks, whereas spurious connections can alter the recovered topology despite affecting few pixels. Road-network extraction is a prominent example of this challenge. RoadTracer [1] addressed this limitation by constructing road graphs through iterative, image-guided exploration rather than relying on raster segmentation followed by heuristic vectorization. VecRoad [19] likewise used segmentation cues to guide iterative graph exploration, while SAM-Road [8] combined road and intersection predictions with learned graph connectivity. More recently, SAM-Road++ [22] improved graph extraction through node-guided resampling and an occlusion-oriented extension strategy. These methods target explicit road-graph construction, whereas our setting retains the raster output of an existing segmenter and corrects its structural errors without modifying the underlying prediction pipeline.

Among segmentation-based methods, Batra et al. [2] jointly learned road orientation and segmentation and introduced a connectivity-refinement network pretrained on corrupted ground-truth masks and subsequently fine-tuned on predicted masks. JTFN [5] coupled topology-preserving and feature-refinement streams through iterative feedback for curvilinear structures, including roads. Unlike these task-specific refinement formulations, we study whether a single correction model can transfer across substantially different frozen segmentation predictors, including those used for road and crack segmentation.

2.2. Topology-Aware Segmentation

Conventional overlap losses are insensitive to the disproportionate structural impact of a few critical pixels. Hu et al. [12] incorporated persistent-homology-based topology constraints into segmentation training. clDice [17] introduced centerline-based supervision for connected tubular structures; Homotopy Warping [10] identified topologically critical pixels using digital topology; and Betti Matching [18] used induced matchings of persistence features to penalize spatially inconsistent topological structures. CAPE [6] introduced a connectivity-aware path-enforcement loss that leverages graph representations and shortest-path optimization to penalize missing and spurious connections in curvilinear structures. Recent work such as SCNP [20] improves topological accuracy by penalizing poorly classified neighboring pixels during network training.

These approaches primarily influence predictions by changing the segmentation training objective. TopoRefiner is complementary: it leaves the upstream predictor frozen and learns a correction in a separate structural representation. Specifically, its encoder is trained to distinguish connectivity-altering perturbations from approximately topology-preserving geometric changes, providing a topology-sensitive target for downstream refinement rather than only an additional loss on the upstream segmentation logits.

2.3. Model-Agnostic Segmentation Refinement

A separate line of work improves existing segmentation outputs without retraining the original predictor. CascadePSP [4] performs class-agnostic, coarse-to-fine refinement with an emphasis on high-resolution boundaries, and SegFix [23] corrects uncertain boundary labels using correspondences to more reliable interior pixels. Post-DAE [13] learns a manifold of plausible segmentation masks through

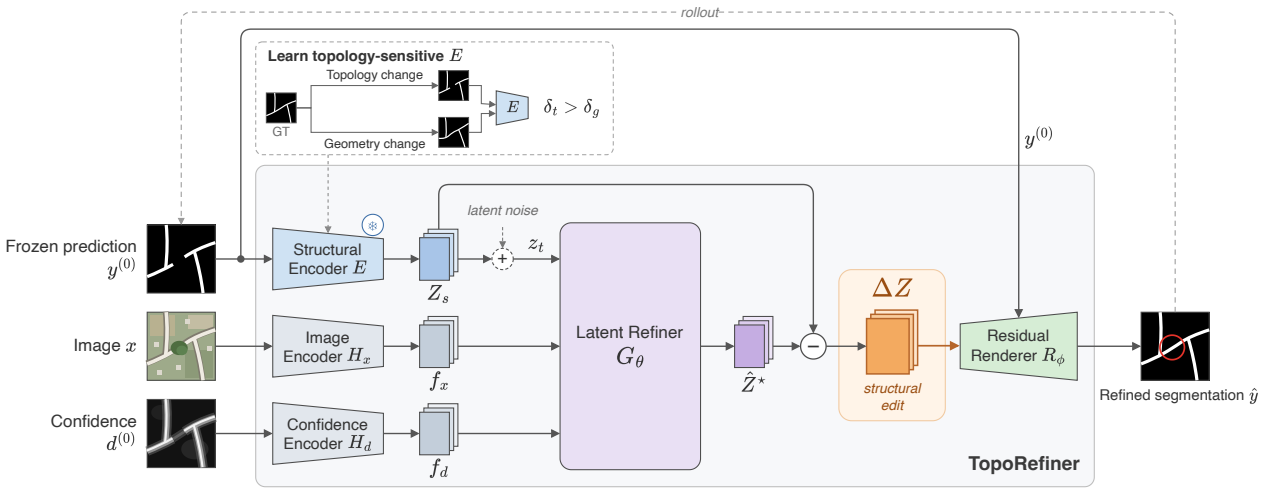


Figure 1. **Overview of TopoRefiner.** A topology-sensitive encoder is first learned from structural and geometric perturbations. During refinement, the prediction, image, and confidence map guide a latent structural correction ΔZ, which is rendered back into segmentation space. Rollout variants recursively reuse the refined prediction.

denoising autoencoders and projects erroneous predictions toward this manifold. SegRefiner [21] casts mask correction as conditional generation with an iterative discrete diffusion process.

More directly related to topology correction, Universal Topology Refinement (UTR) [14] trains a model-agnostic post-processor using synthetic topological perturbations generated from orthogonal polynomial bases. While this strategy broadens the range of training corruptions, refinement operates on segmentation predictions without access to the underlying image evidence. Consequently, the model must infer structural corrections from mask information alone, limiting its ability to distinguish genuine connections from visually unsupported ones. Furthermore, reliance on synthetically generated errors does not necessarily capture the heterogeneous failure modes of real segmentation architectures.

More recently, Curvi-Tracker [7] addresses connectivity errors in curvilinear structures through explicit image-guided tracing. Rather than learning corrections directly in segmentation space, it progressively reconstructs missing structures using local direction and width predictions conditioned on image evidence. This approach demonstrates the value of visually guided post-hoc topology correction, but relies on sequential tracking decisions rather than learning a shared structural correction operator from heterogeneous upstream predictions.

In contrast, TopoRefiner learns a topology-sensitive latent representation that explicitly distinguishes structural changes from approximately topology-preserving geometric variations. Rather than relying solely on synthetic mask perturbations, it learns corrections from both genuine errors of frozen upstream predictors and controlled structural corruptions, conditioning refinement on the source structure, RGB imagery, and upstream confidence. The predicted latent edit is rendered into a pixel-space residual in a single forward pass, without iterative reverse diffusion or explicit path tracing. Crucially, we investigate whether a single refiner, trained jointly across heterogeneous frozen predictors within each dataset, can learn a reusable correction operator without predictor-specific conditioning. We evaluate this hypothesis across road and crack segmentation benchmarks, including composition with existing post-hoc refiners and localized analysis of genuine structural disconnections.

3. Method

Let x denote an input image or volume, $y^{(0)}$ the binary prediction of a frozen segmentation model, and $d^{(0)}$ its continuous confidence field. Our goal is to learn a post-hoc transformation

$$\hat{y} = \mathcal{T}\left(x,\, y^{(0)},\, d^{(0)}\right) \tag{1}$$

that corrects structural errors in $y^{(0)}$ without modifying or retraining the upstream predictor. The central idea of TopoRefiner is to perform this correction through a latent representation that is explicitly sensitive to topology. Instead of directly predicting missing or erroneous pixels, we first encode the source segmentation into a structural latent space, estimate how that representation should change to match the correct topology, and then render the predicted latent edit back into segmentation space. Image evidence and upstream confidence guide the correction, while the original predictor remains completely frozen. TopoRefiner is

trained in two stages. We first learn a structural encoder E whose latent geometry emphasizes connectivity-changing modifications over approximately topology-preserving geometric variation. We then freeze E and train a latent refiner together with a residual renderer to transform imperfect source segmentations toward their ground-truth structural representations. We formulate the method for spatial dimension $D \in \{2, 3\}$, covering both 2D images and 3D volumes. The formulation is identical in both cases; only the corresponding convolutional operators are replaced by their volumetric counterparts. Figure 1 illustrates the overall TopoRefiner framework.

3.1. Topology-Sensitive Structural Representation

Pixel-space distance is poorly aligned with structural error. Removing only a few foreground pixels can disconnect an otherwise continuous structure, whereas substantially changing its width or boundary may leave its topology unchanged. We therefore learn a representation in which these two types of variation are explicitly separated.

For a binary segmentation

$$y \in \{0, 1\}^{C^D}, \tag{2}$$

the structural encoder produces

$$Z = E(y) \in \mathbb{R}^{64 \times (C/16)^D}. \tag{3}$$

The encoder is learned exclusively from ground-truth segmentations and is therefore independent of any particular upstream predictor.

For each ground-truth mask gt, we construct two views. A topology-changing transformation produces m_t, while an approximately topology-preserving geometric transformation produces m_g. The former includes cuts, branch removal, false branches, false components, and false bridges. The latter changes properties such as width, boundary position, local shape, or spatial alignment while preserving the underlying connectivity whenever possible.

We measure the corresponding representation changes as

$$\delta_t = D_z(E(gt), E(m_t)), \qquad \delta_g = D_z(E(gt), E(m_g)), \tag{4}$$

where D_z denotes mean squared distance between corresponding normalized latent features. We require topology-changing transformations to induce a larger displacement than geometry-only transformations:

$$\mathcal{L}_{\text{rank}} = \text{ReLU}(\gamma_r + \delta_g - \delta_t), \qquad \gamma_r = 0.30. \tag{5}$$

Topology-changing perturbations additionally vary in severity. Let m_t^{lo} and m_t^{hi} denote weaker and stronger structural perturbations of the same target, with corresponding distances δ_t^{lo} and δ_t^{hi}. We impose

$$\mathcal{L}_{\text{sev}} = \text{ReLU}\left(\gamma_s + \delta_t^{\text{lo}} - \delta_t^{\text{hi}}\right), \qquad \gamma_s = 0.05, \tag{6}$$

encouraging stronger structural corruption to produce a correspondingly larger latent displacement.

Finally, we prevent the encoder from satisfying these constraints through a collapsed representation using

$$\mathcal{L}_{\text{var}} = 0.5\ \text{ReLU}\left(1 - 10\ \text{Std}(E(gt))\right). \tag{7}$$

The complete representation-learning objective is

$$\mathcal{L}_E = \mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{sev}} + \mathcal{L}_{\text{var}}. \tag{8}$$

After this stage, E is frozen. Importantly, the resulting representation is not trained to segment an image; its role is to provide a structural coordinate system in which topology-altering errors become easier to identify and correct.

3.2. Learning Structural Correction

Given an imperfect source segmentation y_s and its ground truth gt, TopoRefiner learns to transform the structural representation of y_s toward that of gt. The correction is conditioned not only on the source structure, but also on the underlying image and on the confidence of the frozen upstream predictor.

Training sources. Training only from the raw outputs of a segmentation model would expose the refiner to a limited and highly model-dependent error distribution. We therefore construct source masks from a mixture of observed and controlled errors. At each iteration, a source is sampled from

$$\begin{aligned} p_{\text{ord}} = 0.20, \quad p_{\text{quiet}} = 0.25, \quad p_{\text{fp}} = 0.20, \\ p_{\text{ident}} = 0.10, \quad p_{\text{miss}} = 0.15, \quad p_{\text{gap}} = 0.10. \end{aligned} \tag{9}$$

Ordinary samples use the original frozen-model prediction. Quiet samples contain predictions with only minor errors and teach the model to avoid unnecessary edits. False-positive samples introduce spurious branches, components, or bridges. Identity samples use already-correct structures and explicitly encourage preservation. Missing-structure samples remove valid foreground structures synthetically, while prediction-derived gap samples expose the model to disconnections produced by the frozen upstream predictor itself. This mixture is important: TopoRefiner is trained not merely to complete missing connections, but to determine when a structural edit is warranted and what form that edit should take.

Source and target structure. The frozen structural encoder maps the source and target masks to

$$Z_s = E(y_s), \qquad Z^\star = E(gt). \tag{10}$$

The latent channels are standardized using statistics computed from the training set, yielding z_s and $z^\star$.

We perturb the source representation during training using a cosine noise schedule [9, 15]:

$$z_t = \sqrt{\bar{\alpha}_t}\, z_s + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{11}$$

where $t \in \{0, \ldots, T\}$ and $T = 100$. This perturbation prevents the correction network from overfitting to a narrow source-latent distribution and encourages robustness to variation in upstream predictions.

Although Eq. (11) resembles diffusion-style corruption, TopoRefiner is not a reverse-diffusion model. The network directly predicts the corrected structural representation; it never predicts noise and no reverse sampling chain is performed.

Image and confidence evidence. Topology cannot always be inferred from the binary mask alone. A gap may correspond to a missing structure that should be restored or to a genuine separation that should remain disconnected. We therefore condition the correction on two complementary evidence streams.

An image encoder H_x extracts appearance features,

$$f_x = H_x(x), \tag{12}$$

while a confidence encoder H_d processes the continuous output of the frozen upstream predictor,

$$f_d = H_d(d^{(0)}). \tag{13}$$

Both are represented at the same spatial resolution as the structural latent:

$$f_x, f_d \in \mathbb{R}^{64 \times (C/16)^D}. \tag{14}$$

Because different upstream systems expose different score conventions, their outputs are converted to a common continuous confidence representation using fixed score adapters determined from training data. For methods that provide only binary predictions, a continuous field is derived from distance to the predicted boundary while preserving the original segmentation under thresholding. This gives TopoRefiner a consistent confidence interface without requiring model-specific learned heads.

Latent correction. The latent refiner G_θ receives the perturbed source structure together with image and confidence evidence and directly predicts the target structural representation:

$$\hat{z}^\star = G_\theta\left(z_t,\, f_x,\, f_d,\, t/T\right). \tag{15}$$

G_θ is implemented as a lightweight residual convolutional network operating entirely at the latent resolution.

The predicted representation is converted back to the original latent scale, producing $\hat{Z}^\star$. Rather than rendering this representation independently, we explicitly construct the structural edit

$$\Delta Z = \hat{Z}^\star - Z_s. \tag{16}$$

This residual formulation is deliberate: the renderer receives what should change in structural space rather than being asked to reconstruct the entire segmentation from scratch.

Rendering the structural edit. A residual renderer R_ϕ maps ΔZ back into pixel or voxel space. The source segmentation y_s is encoded by a U-Net-style encoder, while the predicted structural edit is projected and injected at the $1/16$-resolution bottleneck. Multi-scale skip connections preserve the geometry already present in the source prediction, and the decoder produces the corresponding pixel-space correction, yielding the refined segmentation $\hat{y}$.

The renderer therefore separates two roles: G_θ determines the desired structural change, whereas R_ϕ realizes that change at the spatial resolution of the segmentation.

3.3. Training Objective

TopoRefiner is supervised simultaneously in structural latent space and segmentation space. The latent prediction is trained toward the encoded ground truth using

$$\mathcal{L}_{\text{latent}} = \|\hat{z}^\star - z^\star\|_2^2. \tag{17}$$

For sufficiently low noise levels, we additionally supervise the rendered segmentation with

$$\mathcal{L}_{\text{seg}} = \frac{1}{2}\left(\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{softDice}}\right) + \mathcal{L}_{\text{clDice}}. \tag{18}$$

The pixel-space loss is applied only when the structural source remains sufficiently informative to support meaningful rendering. The complete objective is therefore

$$\mathcal{L}_{\text{TopoRefiner}} = \mathcal{L}_{\text{latent}} + \mathbf{1}[t \le 40]\, \mathcal{L}_{\text{seg}}. \tag{19}$$

During this stage, the topology-sensitive encoder E remains frozen, while the image encoder H_x, confidence encoder H_d, latent refiner G_θ, and residual renderer R_ϕ are optimized for structural correction.

3.4. Specialist, Shared, and Rollout Variants

The formulation above is independent of how training examples are distributed across upstream predictors. We use this property to instantiate four variants of the same TopoRefiner framework.

Specialist TopoRefiner (S). A separate model is trained for each frozen upstream predictor. This setting allows the correction model to specialize to the characteristic error distribution of a particular segmentation system.

Shared TopoRefiner (J). A single model is trained across heterogeneous frozen upstream predictors. We first sample an upstream system uniformly and then sample a source type according to Eq. (9). The refiner receives no predictor identity, architecture embedding, or model-specific feature representation. Consequently, any correction strategy that it learns must be inferred from the prediction, confidence field, and image evidence themselves.

S and J use the same architecture and objective; they differ only in the distribution of upstream predictions encountered during training.

Rollout-supervised variants. A single correction may not resolve every structural error, particularly when one edit exposes another. We therefore derive rollout-supervised counterparts, denoted **S-RO** and **J-RO**, from the corresponding one-step models. During rollout fine-tuning, refined predictions are recursively reintroduced as source segmentations, allowing the model to learn how its own corrections alter the input distribution. Intermediate states are detached between refinement steps, preventing gradients from propagating through the entire trajectory. The rollout models are evaluated recursively for ten refinement steps. Importantly, S-RO and J-RO are separately rollout-supervised models; they are not obtained by simply applying the one-step S or J checkpoints multiple times.

3.5. Inference

At inference, the upstream segmentation model is evaluated once and remains frozen. Standard S and J models apply a single TopoRefiner pass,

$$y^{(0)} \longrightarrow \hat{y}^{(1)}, \tag{20}$$

whereas rollout-supervised variants recursively update

$$y^{(k+1)} = \mathcal{T}\left(x,\, y^{(k)},\, d^{(k)}\right) \tag{21}$$

for ten refinement steps. Large 2D images and 3D volumes are processed using overlapping windows. We retain continuous output logits for each window, average predictions over overlapping regions, and apply the final threshold only after aggregation. This avoids introducing artificial discontinuities at window boundaries. The same procedure is used in 2D and 3D. For volumetric data, the structural encoder, evidence encoders, latent refiner, and renderer use their 3D convolutional counterparts; the topology-sensitive latent formulation, correction objective, and specialist/shared/rollout training strategies remain unchanged.

4. Experiments

We evaluate TopoRefiner across five benchmarks spanning 2-D road extraction and 3-D neuronal microscopy. All upstream predictors are frozen; TopoRefiner is applied strictly post hoc. We evaluate all four variants introduced in Sec. 3.4: specialist (**S**), shared (**J**), specialist rollout (**S-RO**), and shared rollout (**J-RO**).

4.1. Datasets

2D roads. We evaluate on three standard road-segmentation benchmarks: Massachusetts Roads ($1{,}108/14/49$ train/val/test images, 1500^2), DeepGlobe ($4{,}358/120/1{,}245$ labelled 1024^2 tiles), and RoadTracer ($148/32/60$ city-disjoint 4096^2 tiles).

3D neuronal microscopy. FlyLight 3D contains 40 isotropic light-microscopy stacks of fly optic-lobe neurons with extremely sparse tubular annotations (0.11% foreground), split into 32/3/5 train/val/test stacks. NeuroFly 3D contains 123 independent 128^3 neuronal crops from three whole-brain imaging sources, with 0.17% foreground and an 89/17/17 train/val/test split. Both datasets use one-voxel-dilated skeleton annotations and are evaluated using overlapping 96^3 windows.

4.2. Evaluation Protocol and Metrics

For 2D experiments, U-Net provides the common frozen base prediction; for 3D experiments, we use nnU-Net. We additionally evaluate TopoRefiner after existing topology-oriented systems, including Batra connectivity refinement [3], UTR [14], Curvi-Tracker [7], Betti Matching [18], and Homotopy Warping [11]. This tests not only correction of a conventional segmenter, but also whether TopoRefiner can provide additional structural improvement after methods that already explicitly target topology.

We report Dice for region overlap and clDice for centerline agreement. Structural accuracy is measured using absolute errors in β_0 and β_1, corresponding to connected components and cycles, respectively, together with Betti Matching error (BettiM).

Aggregate topological error alone does not establish that a particular disconnection has been repaired. We therefore additionally report CONNECT, the percentage of a fixed set of base-model disconnections that are successfully restored. The same frozen base-model gap set is used for every method within a dataset, containing 1,936, 15,626, 28,148, 55, and 67 cases for Massachusetts, DeepGlobe, RoadTracer, FlyLight, and NeuroFly, respectively. All hyperparameters and checkpoints are selected using validation data.

Computational cost. TopoRefiner remains lightweight in both settings. The 2D model used across the road benchmarks contains 5.71M parameters, while the volumetric 3D model used for FlyLight and NeuroFly contains 16.85M parameters. These counts exclude the frozen upstream seg-

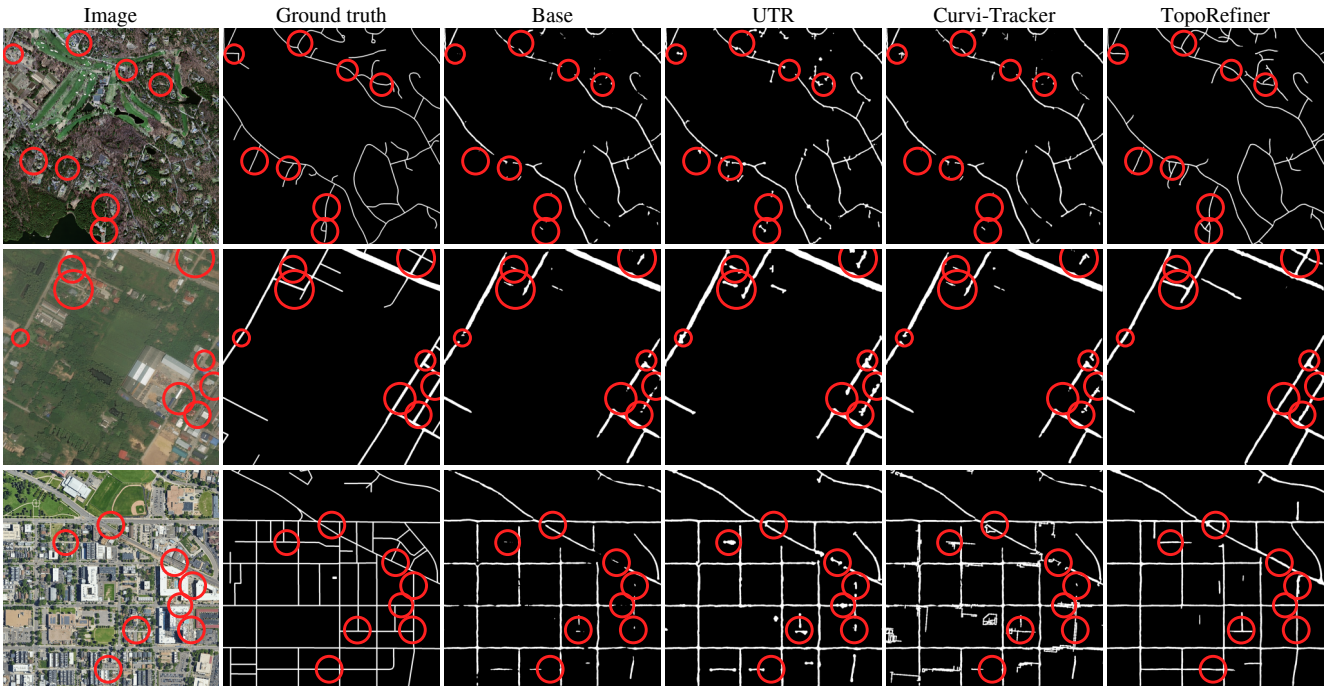


Figure 2. **Qualitative results on 2-D road segmentation.** From left to right: input image, ground truth, frozen U-Net prediction, UTR, Curvi-Tracker, and TopoRefiner. Red circles highlight representative structural errors. TopoRefiner restores missing road connections that remain unresolved by the base prediction and competing refiners.

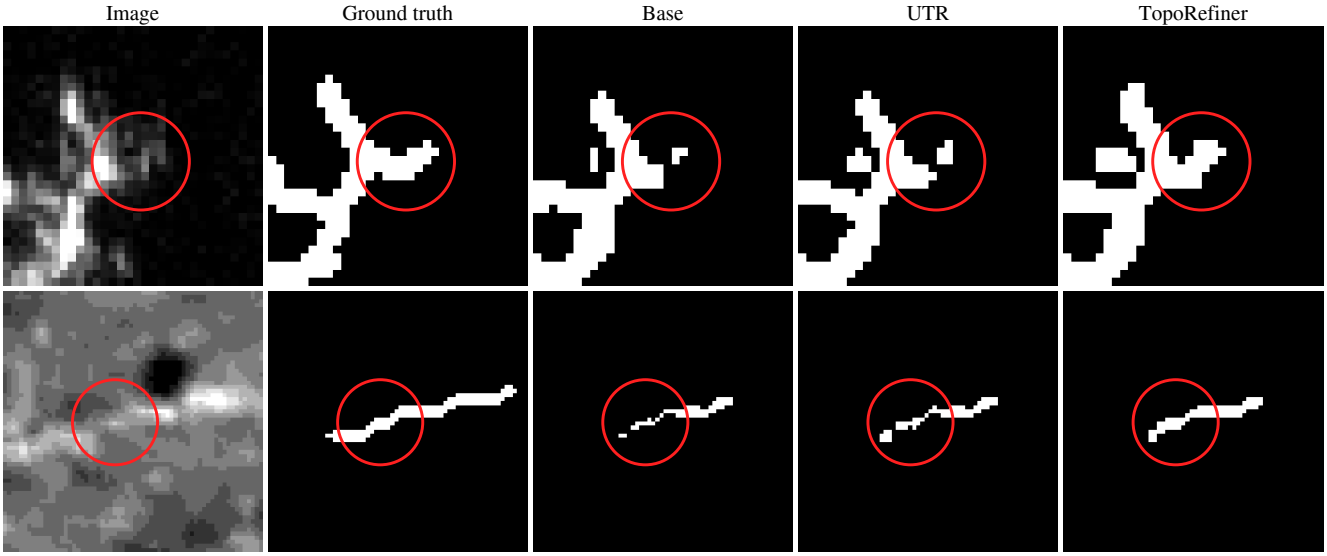


Figure 3. **Qualitative results on 3-D neuronal segmentation.** From left to right: image, ground truth, frozen nnU-Net prediction, UTR-3D, and TopoRefiner. Red circles highlight fragmented neuronal structures. TopoRefiner recovers missing connections that remain disconnected in the upstream and UTR-3D predictions.

menter, and the specialist and shared variants use identical architectures and parameter counts.

4.3. Quantitative Results

Comparison with topology-oriented refiners. Table 1 compares TopoRefiner with existing post-hoc topology-refinement methods on the common frozen base predic-

tions. Across all three road benchmarks, TopoRefiner provides the strongest overall combination of segmentation quality, topological accuracy, and direct connectivity restoration. In particular, the specialist variants substantially improve Dice and clDice, while rollout supervision yields markedly stronger reductions in connected-component and Betti errors. The CONNECT evaluation confirms that these improvements correspond to actual recovery of missing connections rather than merely changes in global topology statistics. The same trend extends to 3-D: despite already strong nnU-Net predictions and extremely sparse neuronal foregrounds, TopoRefiner improves several structural metrics and achieves successful reconnections that are not obtained by UTR-3D on the same frozen gaps.

Refinement across heterogeneous upstream systems. Tabs. 2 to 6 evaluates TopoRefiner on top of a broader set of upstream predictions, including topology-aware losses and existing refinement methods. TopoRefiner continues to provide substantial structural gains after Batra CR, UTR, Curvi-Tracker, Betti Matching, and Homotopy Warping, showing that its effect is complementary rather than redundant with these approaches. This behavior is particularly important: TopoRefiner does not require the upstream prediction to originate from a conventional segmenter, and remains effective even when topology has already been explicitly addressed during training or post-processing.

Choosing a TopoRefiner variant. The four variants expose a clear and useful accuracy–topology trade-off. When preserving Dice and geometric fidelity is the main objective, TopoRefiner-S is the strongest default, consistently providing the most conservative corrections while still improving structure. When connectivity and connected-component accuracy are the priority, S-RO is generally preferable: repeated, rollout-supervised refinement produces the strongest reductions in β_0 and frequently the best overall Betti error, particularly on fragmented road networks. For applications requiring a single refiner across heterogeneous upstream models, J provides the more conservative shared operating point, whereas J-RO favors stronger topological correction and is particularly effective for cycle-related errors. Thus, TopoRefiner provides not a single fixed operating point, but a practical choice between overlap preservation, aggressive topology correction, and cross-predictor reuse.

4.4. Qualitative Results

Figures 2 and 3 visualize representative corrections in 2D and 3D. In the road examples, TopoRefiner restores interrupted road segments along visually supported paths, including cases where UTR and Curvi-Tracker leave the disconnection unresolved. The 3D examples show the same behavior on sparse neuronal structures: TopoRefiner reconnects fragmented branches that remain separated in the nnU-Net and UTR-3D predictions. These examples qualitatively support the quantitative finding that TopoRefiner improves connectivity while largely preserving the surrounding structure.

5. Conclusion

We introduced TopoRefiner, a post-hoc topology-refinement framework that learns structural correction in a topology-sensitive latent space while leaving the upstream segmenter completely frozen. Across 2D road segmentation and 3D neuronal microscopy, TopoRefiner consistently improves structural quality over heterogeneous upstream systems, including topology-aware predictors and existing refiners. The specialist, shared, and rollout-supervised variants expose a practical trade-off between overlap preservation, stronger topological correction, and cross-predictor reuse, allowing the refinement strategy to be matched to the application objective. These results support a broader view of topology refinement as a reusable post-hoc operation rather than a constraint that must be built into a particular segmentation model. Future work will investigate dataset-agnostic topology refinement, aiming to generalize a single refiner not only across upstream predictors, but also across datasets and structural domains without retraining.

Table 1. **Comparison with topology-oriented refinement methods across five datasets.** For 2-D datasets, methods refine a frozen U-Net prediction; for 3-D datasets, they refine nnU-Net. CONNECT (%) measures successful reconnection on the same frozen base-model gap set for each dataset. Higher Dice, clDice, and CONNECT are better; lower Betti errors and BettiM are better.

Method	Dice↑	clDice↑	CONNECT↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
Massachusetts Roads						
U-Net	.6903	.8398	0.00	30.08	46.94	85.76
U-Net + Batra CR	.7440	.8445	5.99	76.57	32.08	129.10
U-Net + UTR	.6979	.8487	23.76	12.82	32.86	56.20
U-Net + Curvi-Tracker	.6864	.8326	6.46	23.57	38.59	89.59
U-Net + TopoRefiner-S	**.7523**	**.8616**	33.68	13.71	25.67	61.31
U-Net + TopoRefiner-J	.7451	.8536	13.79	15.88	39.45	65.86
U-Net + TopoRefiner-S-RO	.7317	.8408	**45.04**	**3.31**	**18.29**	**55.31**
U-Net + TopoRefiner-J-RO	.7440	.8529	26.39	6.90	31.67	57.27
DeepGlobe						
U-Net	.5773	.6540	0.00	19.62	8.03	29.03
U-Net + Batra CR	.5974	.6656	6.49	41.24	6.89	55.51
U-Net + UTR	.6013	.6959	22.46	7.83	7.01	16.21
U-Net + Curvi-Tracker	.5881	.6626	11.41	13.00	7.32	28.98
U-Net + TopoRefiner-S	**.6594**	**.7634**	**54.54**	4.88	4.84	13.15
U-Net + TopoRefiner-J	.6321	.7360	38.67	6.90	6.31	15.35
U-Net + TopoRefiner-S-RO	.6435	.7410	53.23	**2.48**	**4.23**	**11.33**
U-Net + TopoRefiner-J-RO	.6271	.7298	46.74	3.13	5.71	12.20
RoadTracer						
U-Net	.3797	.4296	0.00	695.03	446.17	1181.93
U-Net + Batra CR	.4159	.4700	7.09	949.82	334.07	1526.35
U-Net + UTR	.4465	.5273	24.90	287.03	401.67	787.77
U-Net + Curvi-Tracker	.3731	.3978	17.04	262.22	1695.55	2791.40
U-Net + TopoRefiner-S	**.4861**	.5670	42.19	177.33	**280.12**	759.22
U-Net + TopoRefiner-J	.4273	.4865	20.58	278.90	419.03	768.10
U-Net + TopoRefiner-S-RO	.4815	**.5711**	**52.38**	78.15	303.35	**605.77**
U-Net + TopoRefiner-J-RO	.4490	.5185	37.57	**67.57**	301.80	634.00
FlyLight 3-D						
nnU-Net	**.7548**	.9116	0.00	2.80	4.00	21.60
nnU-Net + UTR-3D	.7435	.9102	0.00	2.40	3.80	21.20
nnU-Net + TopoRefiner-S	.7502	.9133	0.00	1.40	4.40	**21.00**
nnU-Net + TopoRefiner-J	.7486	**.9149**	0.00	1.80	3.40	22.40
nnU-Net + TopoRefiner-S-RO	.7410	.9045	0.00	**.60**	**3.20**	23.40
nnU-Net + TopoRefiner-J-RO	.7406	.9040	**3.64**	1.00	3.40	22.80
NeuroFly 3-D						
nnU-Net	**.6732**	.8110	0.00	5.47	1.06	7.71
nnU-Net + UTR-3D	.6671	.8108	0.00	5.47	.94	7.53
nnU-Net + TopoRefiner-S	.6701	**.8189**	**5.97**	5.35	1.06	7.47
nnU-Net + TopoRefiner-J	.6685	.8150	2.99	**4.88**	.94	**7.18**
nnU-Net + TopoRefiner-S-RO	.6649	.8066	2.99	5.65	.88	7.71
nnU-Net + TopoRefiner-J-RO	.6564	.8002	4.48	5.06	**.82**	7.29

CONNECT is the percentage of frozen base-model gap windows for which refinement reduces the number of connected components while recovering at least 50% of the gap midline. All methods are evaluated on the same fixed gap set within each dataset. TopoRefiner CONNECT values use the window-level evaluation employed during gap testing; competing refiners are evaluated from their frozen full-image test predictions.

Table 2. **TopoRefiner across representative upstream methods on Massachusetts Roads.**

Upstream	**Refinement**	Dice↑	clDice↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
U-Net	Base	.6903	.8398	30.08	46.94	85.76
	+ TopoRefiner-S	**.7523**	**.8616**	13.71	25.67	61.31
	+ TopoRefiner-J	.7451	.8536	15.88	39.45	65.86
	+ TopoRefiner-S-RO	.7317	.8408	**3.31**	**18.29**	**55.31**
	+ TopoRefiner-J-RO	.7440	.8529	6.90	31.67	57.27
Batra CR	U-Net + Batra CR	.7440	.8445	76.57	32.08	129.10
	+ TopoRefiner-S	.7527	.8612	23.59	30.45	68.82
	+ TopoRefiner-J	**.7556**	.8602	29.08	31.39	78.27
	+ TopoRefiner-S-RO	.7193	.8498	**4.76**	**24.00**	**49.69**
	+ TopoRefiner-J-RO	.7542	**.8615**	12.57	29.71	60.61
Betti Matching	U-Net + Betti Matching Loss	.5105	.8494	25.80	26.88	62.59
	+ TopoRefiner-S	**.7413**	.8595	18.12	**18.12**	66.33
	+ TopoRefiner-J	.7157	**.8601**	13.49	25.35	50.96
	+ TopoRefiner-S-RO	.7382	.8502	17.88	27.78	71.94
	+ TopoRefiner-J-RO	.7347	.8483	**5.31**	24.69	**49.39**
UTR	U-Net + UTR	.6979	.8487	12.82	32.86	56.20
	+ TopoRefiner-S	.7402	.8502	11.02	28.27	59.82
	+ TopoRefiner-J	**.7421**	**.8547**	10.12	30.39	54.67
	+ TopoRefiner-S-RO	.7358	.8420	12.24	39.35	65.47
	+ TopoRefiner-J-RO	.7420	.8504	**5.22**	**25.84**	**53.80**
Curvi-Tracker	U-Net + Curvi-Tracker	.6864	.8326	23.57	38.59	89.59
	+ TopoRefiner-S	**.7410**	**.8474**	18.61	35.06	73.10
	+ TopoRefiner-J†	.7302	.8404	14.92	36.47	76.29
	+ TopoRefiner-S-RO	.7271	.8337	14.92	36.82	**69.41**
	+ TopoRefiner-J†-RO	.7255	.8381	**8.47**	**29.57**	74.90
Homotopy Warping	U-Net + Homotopy Warping Loss	.7730	.8717	32.12	27.22	72.65
	+ TopoRefiner-S	.7736	.8733	20.04	27.88	59.96
	+ TopoRefiner-J	**.7791**	**.8793**	18.45	25.86	58.92
	+ TopoRefiner-S-RO	.7512	.8593	**8.49**	**20.49**	55.71
	+ TopoRefiner-J-RO	.7726	.8757	8.88	23.22	**51.82**

†Joint model retrained with Curvi-Tracker included in the heterogeneous training pool.

Table 3. **TopoRefiner across representative upstream methods on DeepGlobe.**

Upstream	**Refinement**	Dice↑	clDice↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
U-Net	Base	.5773	.6540	19.62	8.03	29.03
	+ TopoRefiner-S	**.6594**	**.7634**	4.88	4.84	13.15
	+ TopoRefiner-J	.6321	.7360	6.90	6.31	15.35
	+ TopoRefiner-S-RO	.6435	.7410	**2.48**	**4.23**	**11.33**
	+ TopoRefiner-J-RO	.6271	.7298	3.13	5.71	12.20
Batra CR	U-Net + Batra CR	.5974	.6656	41.24	6.89	55.51
	+ TopoRefiner-S	**.6356**	.7241	10.83	6.28	19.98
	+ TopoRefiner-J	.6325	.7260	13.29	5.84	23.73
	+ TopoRefiner-S-RO	.6268	**.7329**	**2.76**	**4.45**	**11.45**
	+ TopoRefiner-J-RO	.6309	.7307	5.76	5.23	15.89
Betti Matching	U-Net + Betti Matching Loss	.6002	.7230	16.19	6.28	25.08
	+ TopoRefiner-S	**.6680**	**.7745**	4.96	4.67	12.28
	+ TopoRefiner-J	.6260	.7632	6.60	5.24	14.28
	+ TopoRefiner-S-RO	.6444	.7589	**1.94**	**4.13**	**9.73**
	+ TopoRefiner-J-RO	.6345	.7550	3.23	4.35	11.25
UTR	U-Net + UTR	.6013	.6959	7.83	7.01	16.21
	+ TopoRefiner-S	.6355	.7408	5.93	5.11	14.02
	+ TopoRefiner-J	.6227	.7351	4.89	5.86	13.07
	+ TopoRefiner-S-RO	**.6453**	**.7427**	2.50	4.64	10.89
	+ TopoRefiner-J-RO	.6290	.7369	**2.07**	**4.58**	**10.61**
Curvi-Tracker	U-Net + Curvi-Tracker	.5881	.6626	13.00	7.32	28.98
	+ TopoRefiner-S	.6327	.7213	7.11	6.42	16.97
	+ TopoRefiner-J†	.6298	.7218	6.30	5.76	17.94
	+ TopoRefiner-S-RO	**.6386**	**.7370**	3.83	5.24	**13.07**
	+ TopoRefiner-J†-RO	.6302	.7280	**2.69**	**4.86**	13.78
Homotopy Warping	U-Net + Homotopy Warping Loss	.6787	.7423	26.81	5.87	35.40
	+ TopoRefiner-S	**.6896**	**.7627**	13.97	5.11	21.71
	+ TopoRefiner-J	.6723	.7571	13.56	4.79	21.74
	+ TopoRefiner-S-RO	.6737	.7537	**5.39**	4.27	**14.41**
	+ TopoRefiner-J-RO	.6658	.7501	7.89	**4.20**	16.38

†Joint model retrained with Curvi-Tracker included in the heterogeneous training pool.

Table 4. **TopoRefiner across representative upstream methods on RoadTracer.**

Upstream	**Refinement**	Dice↑	clDice↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
U-Net	Base	.3797	.4296	695.03	446.17	1181.93
	+ TopoRefiner-S	**.4861**	.5670	177.33	**280.12**	759.22
	+ TopoRefiner-J	.4273	.4865	278.90	419.03	768.10
	+ TopoRefiner-S-RO	.4815	**.5711**	78.15	303.35	**605.77**
	+ TopoRefiner-J-RO	.4490	.5185	**67.57**	301.80	634.00
Batra CR	U-Net + Batra CR	.4159	.4700	949.82	334.07	1526.35
	+ TopoRefiner-S	**.4815**	.5613	276.05	346.58	802.43
	+ TopoRefiner-J	.4651	.5307	286.02	380.52	795.00
	+ TopoRefiner-S-RO	.4682	**.5793**	**29.78**	280.13	**552.78**
	+ TopoRefiner-J-RO	.4759	.5536	75.12	**261.65**	670.67
Betti Matching	U-Net + Betti Matching Loss	.4160	.6020	749.87	**183.32**	1356.65
	+ TopoRefiner-S	**.5024**	.5830	273.30	195.88	889.25
	+ TopoRefiner-J	.4948	**.6053**	227.93	233.93	767.83
	+ TopoRefiner-S-RO	.4890	.5679	191.45	248.48	1072.50
	+ TopoRefiner-J-RO	.4770	.5602	**61.20**	200.35	**657.62**
UTR	U-Net + UTR	.4465	.5273	287.03	401.67	787.77
	+ TopoRefiner-S	**.4780**	.5580	178.27	360.18	690.05
	+ TopoRefiner-J	.4721	.5548	142.65	376.83	639.68
	+ TopoRefiner-S-RO	.4647	.5476	76.18	292.17	**628.22**
	+ TopoRefiner-J-RO	.4763	**.5589**	**56.07**	**250.42**	646.25
Curvi-Tracker	U-Net + Curvi-Tracker	.3731	.3978	262.22	1695.55	2791.40
	+ TopoRefiner-S	.4535	.5048	403.05	264.28	1279.57
	+ TopoRefiner-J†	.4304	.4805	340.17	499.37	1461.47
	+ TopoRefiner-S-RO	**.4668**	**.5314**	151.73	**220.23**	878.10
	+ TopoRefiner-J†-RO	.4467	.5233	**57.38**	237.00	**671.85**
Homotopy Warping	U-Net + Homotopy Warping Loss	.5117	.5860	1015.87	373.28	1496.18
	+ TopoRefiner-S	**.5235**	**.6029**	379.17	295.00	900.43
	+ TopoRefiner-J	.5179	.5960	390.50	341.97	872.87
	+ TopoRefiner-S-RO	.4974	.5774	**56.73**	191.32	**725.08**
	+ TopoRefiner-J-RO	.5018	.5889	98.83	**188.98**	731.08

†Joint model retrained with Curvi-Tracker included in the heterogeneous training pool.

Table 5. **TopoRefiner across representative upstream methods on FlyLight 3-D.**

Upstream	**Refinement**	Dice↑	clDice↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
nnU-Net	Base	**.7548**	.9116	2.80	4.00	21.60
	+ TopoRefiner-S	.7502	.9133	1.40	4.40	**21.00**
	+ TopoRefiner-J	.7486	**.9149**	1.80	3.40	22.40
	+ TopoRefiner-S-RO	.7410	.9045	**.60**	**3.20**	23.40
	+ TopoRefiner-J-RO	.7406	.9040	1.00	3.40	22.80
UTR-3D	nnU-Net + UTR-3D	**.7435**	.9102	2.40	3.80	**21.20**
	+ TopoRefiner-S	.7430	.9087	1.80	2.40	23.60
	+ TopoRefiner-J	.7401	**.9116**	1.40	3.20	22.80
	+ TopoRefiner-S-RO	.7233	.8955	**.60**	4.60	24.60
	+ TopoRefiner-J-RO	.7328	.9018	1.60	**2.20**	23.60

Table 6. **TopoRefiner across representative upstream methods on NeuroFly 3-D.**

Upstream	**Refinement**	Dice↑	clDice↑	$\beta_0 \downarrow$	$\beta_1 \downarrow$	BettiM↓
nnU-Net	Base	**.6732**	.8110	5.47	1.06	7.71
	+ TopoRefiner-S	.6701	**.8189**	5.35	1.06	7.47
	+ TopoRefiner-J	.6685	.8150	**4.88**	.94	**7.18**
	+ TopoRefiner-S-RO	.6649	.8066	5.65	.88	7.71
	+ TopoRefiner-J-RO	.6564	.8002	5.06	**.82**	7.29
UTR-3D	nnU-Net + UTR-3D	.6671	.8108	5.47	.94	7.53
	+ TopoRefiner-S	**.6672**	**.8148**	**5.18**	.94	**7.12**
	+ TopoRefiner-J	.6655	.8113	**5.18**	.94	7.24
	+ TopoRefiner-S-RO	.6641	.8049	5.88	.94	7.82
	+ TopoRefiner-J-RO	.6549	.7974	5.53	**.88**	7.47

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